S1 Supplementary Methods

Cohort-anchored export, within-child age analysis, and robustness checks

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S1.1 Measurement and provenance contract

The source is Figshare 5598814.v1 plus associated metadata. This supplement documents the artifact-masked re-export `neuralmanifolddynamics_figshare_infant_eeg_20260805_132612`, which supersedes an earlier export (`..._20260805_102438`) that did not mask upstream BAD_-interval artifact annotations. The ingest pipeline imported 715 publicly released BAD_-interval annotations (2,143.7 s total) and masks flagged samples before epoching; any 8-s epoch overlapping a masked interval is dropped rather than bridged (no artificial time joins across a gap). This reduced the usable epoch pool from 6,929 to 5,265 epochs (median 78.5 per session, range 20–109). The run manifest reports 70 H5 files, 22 TD children, and 6 visit labels. It realizes subject- and cohort-anchored coordinate layers. The frozen subject-balanced IQR anchor is `figshare_infant_eeg_global_cohort_all_visits_iqr_v2_1`, refit on the artifact-masked epoch pool (anchor hash `45bbfa21...c301a89e`). TD25v4 has no BAD_ annotation available but was already excluded from this export by the coverage rule, so the absence of an annotation does not affect any analysed session.

S1.2 Session-level NDT summaries

Each H5 provides 3D MNPS, 9D MNPS, and windowed 3D/9D MNJs. For every accepted (`ok` or `adjusted`) recording, `01_export_session_metrics.py` computes medians of cohort and subject coordinates, MNJ invariants, and reachability summaries, then writes one row per visit. The reachability configuration uses the export's cohort-anchored primary layer, k=12, horizon=3, ridge local models, and no additional standardization. Epochs are never entered as independent observations.

S1.3 Within-child age estimator

For child i at visit j, let $a_{i,j}$ be age in days and $y_{i,j}$ an endpoint. We estimate

$$\beta = \sum_{i,j} \left(a_{i,j} - |(a)_i\right) \frac{y_{i,j} - |(y)_i}{\sum_{i,j}} \left(a_{i,j} - |(a)_i\right)^2. \tag{1}$$

The displayed effect is β per 30 days and a within-child standardized slope. Only children with two or more distinct ages contribute. Child-cluster bootstrap intervals resample children; the raw p-value is a two-sided child-sign permutation of centered endpoint trajectories. BH-FDR is reported as a sensitivity analysis, not a confirmatory decision rule (Benjamini & Hochberg, 1995).

S1.4 Primary and robustness families

The cohort-coordinate family (3D m, d, e and the nine 9D coordinate medians) is the primary analysis. Each endpoint is modelled by Gaussian GEE with exchangeable child clustering and a 3-df cubic B-spline of age; the omnibus spline Wald test is FDR-corrected across the 12 coordinates. The within-child MNJ/reachability family (primary 3D and secondary 9D) and the cohort-pooled reachability family (S1.7) are robustness checks on whether the same age association also appears as a local-dynamics effect, reported in full in Supplement S2. No event, stage, regional, or Bayley-outcome inference is performed.

S1.5 Robustness

The direct-age sensitivity retains only age values marked `spreadsheet_direct`; 40 visits are available for the primary cohort-coordinate GEE and 36 visits from 18 children contribute to the within-child robustness family (both reported in Supplement S2, including endpoints that were not previously surfaced in the main text). First-to-last summaries are descriptive because follow-up spans differ by child. Sixty-nine geometry contracts are `ok` and one is `adjusted`; both statuses are accepted by the export invalidity policy.

S1.6 Within/between-child decomposition and cluster-permutation check

Two additional robustness checks were run on the primary cohort-coordinate GEE (`03_cohort_coordinate_age.py`) to separate longitudinal from cross-sectional association and to assess small-cluster inference.

Within/between decomposition. For each endpoint, the raw age spline is replaced by a linear child-mean age term $|(a)_i$ ("between") plus a 3-df cubic B-spline of the centered age $a_{i,j} - |(a)_i$ ("within"), fit in the same GEE. A 3-df Wald test isolates the within-child spline coefficients; a 1-df Wald test isolates the between-child coefficient. The unequal degrees of freedom make this an asymmetric diagnostic and can give the between-child term less flexibility; it is therefore interpreted descriptively rather than as proof that all between-child structure is absent. This is reported as an exploratory diagnostic on the primary family and is not itself FDR-corrected as a second confirmatory family.

Child-cluster row-permutation check (M4). Because only 22 clusters is a known regime for anti-conservative GEE sandwich-covariance inference, age_days is permuted across all 70 visit-rows (breaking the age-outcome association while preserving the design, n, and clustering), the identical spline GEE is refit, and the omnibus Wald chi-squared statistic is recorded; this is repeated 500 times per endpoint with a fixed seed. The permutation p-value is (number of permuted statistics >= observed + 1) / (permutations + 1). BH-FDR is then applied across the 12 permutation p-values themselves, separately from the primary family's own FDR correction, since the two tests answer different questions (asymptotic vs. finite-sample-robust significance).

S1.7 Cohort-pooled reachability estimator

`05_cohort_pooled_reachability.py` estimates a genuinely cohort-level local-dynamics summary, distinct from the per-session within-child family (S1.2-S1.3). For every H5, all valid epoch-to-epoch transitions (epoch $t \rightarrow$ epoch $t + 1$) in the cohort-anchored 3D chart are pooled into a single set of 5,195 transition pairs tagged by child and visit. For each of the 70 visits, the query point is that visit's own cohort m, d, e median (identical to the coordinate reported in S2.1); the neighbor pool excludes every visit belonging to the same child (leave-one-child-out, not merely the same visit), and the $k = 12$ nearest pool transitions by Euclidean distance to the query point are used to fit a ridge local linear model ($\alpha = 10^{-}3$) exactly as in the per-session estimator, then propagated to reachability summaries (log-volume, effective dimension, anisotropy, directional entropy, rotational proxy) with the identical horizon ($= 3$), jitter ($10^{-}6$), and aggregation as the per-session pipeline. The resulting 70 per-visit summaries are tested with the same within-child estimator as S1.3. The median neighbor k -th-distance and the number of distinct contributing children are reported per visit as density diagnostics (S2.8).

S1.8 Conventional-EEG comparator family

`06_conventional_eeg_age.py` extracts 34 `eeg_conventional_*` per-epoch comparator columns from the same H5 recordings' `features_raw/names`/`features_raw/values` layout, drawn from the upstream ingest pipeline's `tier1`, `complexity`, and `connectivity` packs (`config_ingest_figshare_infant_eeg.yaml`): relative delta/theta/alpha/beta/gamma and infant-specific alpha/theta band power, five band-power ratios (alpha/theta, beta/alpha, delta/alpha, theta/alpha, and a slowing index), three spectral peak/edge summaries, three entropy/complexity summaries (permutation entropy, spectral entropy, Hjorth complexity), and 16 alpha/theta coherence summaries (mean and SD across epochs) at four channel pairs (F4-P4, F8-P4, central-parietal, frontal-central). Each visit's endpoint value is the per-epoch median across that session's accepted epochs, computed exactly like the primary MNPS/MNJ session summaries (S1.2). The identical cohort-level spline GEE, within/between decomposition, and child-cluster row-permutation check used for the primary cohort-coordinate family (S1.4, S1.6) are applied to this family, plus the within-child sign-permutation slope estimator (S1.3) used for the local-dynamics families; all four are FDR-corrected across this family's 34 endpoints separately from every other family in this study. An endpoint is reported as `not_testable` (excluded from that family's FDR correction) rather than causing a failure if fewer than 10 sessions, fewer than 4 distinct ages, zero outcome variance, or fewer than 3 children with within-child age variation remain after dropping missing values – this guard specifically matters for the `connectivity` comparators, whose per-recording coverage is materially lower than the other packs (S2.12).

This family uses a second H5 re-extraction (`neuralmanifolddynamics_figshare_infant_eeg_20260806_122539`, one day later than the primary export, with the same 5,265-epoch artifact mask and a fix to the `connectivity` comparator's channel/coverage handling) rather than the primary export used everywhere else in this study. Before use, every one of the 74 shared numeric MNPS/MNJ/reachability session-metric columns (including all 12 primary cohort-coordinate

endpoints) was confirmed bit-identical (max absolute difference 0.0) between the two exports; see `results/README.md` for the full provenance note.

Bibliography

Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x